AXZ OIL AND GAS

Production Performance Review

Weeks 1 – 4 | Four-Location Operational Analysis

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Prepared by: Isaac Okolie | Data Analyst

Client: AXZ Oil and Gas Production Company

Locations: Brass | Ekeremor | Nembe | Southern Ijaw

Tool: Power BI | Dataset: Weekly Employee Production Data

11.02	■898	232	285K
UPH Achieved	CPU Achieved	UPI Achieved	Units Produced
Target: 11 ■	Target: ■750 ■	Target: 250 ■	4 Weeks Total

1. Executive Overview

Four weeks into operations, AXZ Oil and Gas Production Company has produced 285,124 units across four rig locations — Brass, Ekeremor, Nembe, and Southern Ijaw. At face value, the headline numbers appear acceptable. The company is producing units at 11.02 per hour, marginally above its target of 11. But beneath that headline, a more complicated picture emerges.

Two of the three non-negotiable performance targets — cost and quality — are being missed. The cost of producing a single unit has reached ■898 against a ceiling of ■750, representing a 19.75% cost overrun. Quality performance, measured by the number of units produced before a quality issue occurs, stands at 232 against a target of 250. And when the numbers are broken down by location, the performance gap between the best and worst-performing rigs is significant enough to demand immediate management attention.

This report investigates the four-week production dataset, answers the nine questions management raised in the business brief, and delivers a set of targeted recommendations supported by the data — including a finding that may be the single most actionable insight available: staff training hours are strongly correlated with production quality across all rig locations.

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2. Business Background

AXZ Oil and Gas Production Company is a hydrocarbon production operation running four active rig locations across the Niger Delta region — Brass, Ekeremor, Nembe, and Southern Ijaw. Each location operates independently, with its own workforce, production targets, and cost structure.

The company measures operational performance against four targets set by management at the beginning of operations:

Target	Metric	Threshold	Business Logic
Production	UPH — Units Per Hour	≥ 11 units/hour	Minimum output rate required to meet production commitments
Cost	CPU — Cost Per Unit	≤ ■750 per unit	Maximum allowable spend to produce one unit and remain profitable
Quality	UPI — Units Per Issue	≥ 250 units before a defect	Minimum quality reliability standard — fewer than 250 signals a process problem

Target	Metric	Threshold	Business Logic
Training	Training Hours	≥ 1 hour per staff/week	Staff development commitment — expected to improve long-term performance

The operations had been running for four weeks when management commissioned this analysis. With 113 employees distributed across the four rigs, and weekly production data recorded at individual employee level, the dataset provides a granular view of how each location and each worker is performing against these standards.

3. The Business Problem

At the start of operations, management established clear performance targets. Four weeks later, they had production data — but no structured view of how performance was tracking against those targets, which locations were meeting expectations, and crucially, why some locations might be performing better than others.

Without this visibility, several risks were accumulating silently. A location producing units at ■1,000 per unit while the target is ■750 could be operating for weeks without anyone calculating the cost overrun at scale. A rig with a quality failure rate three times worse than another might not trigger concern because no one was comparing the four locations side by side.

There was also a strategic question embedded in the business brief that went beyond simple target monitoring. Management wanted to understand whether there was a relationship between staff training hours and key production metrics. If training was genuinely improving performance, that would be one of the most cost-effective investments the company could make. If it was not, resources might be better directed elsewhere.

The data existed. The questions had been asked. This engagement was commissioned to turn that raw weekly data into answers.

4. Project Objectives

1. Measure overall production output

Calculate total units produced across all four rig locations over the four-week period and compare against operational expectations.

2. Evaluate cost efficiency

Determine the actual Cost Per Unit (CPU) at each location and assess whether production spending is within the ■750 per unit ceiling.

3. Assess quality performance

Calculate the Units Per Issue (UPI) metric at each location and identify which rigs are failing to meet the 250-unit quality reliability standard.

4. Analyze workforce distribution

Determine the number of employees across all locations and evaluate whether staffing levels are consistent with output differences.

5. Examine working hours

Measure average hours worked per employee across all locations and assess the relationship between working hours and production performance.

6. Evaluate training compliance

Determine whether staff at each location are completing the minimum one hour of training per week and identify non-compliant locations.

7. Quantify training impact

Measure the statistical correlation between training hours and key performance metrics — UPH, CPU, and UPI — to determine whether training investment is producing measurable returns.

8. Identify location-level performance gaps

Rank all four locations across all four KPIs to surface the specific locations and metrics requiring the most urgent management attention.

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5. Data Description

The dataset consists of four weekly Excel files — one for each week of operations. Each file records individual employee-level production data for that week, enabling analysis at both the individual and location level.

Property	Detail
Total Records (combined)	452 employee-week records (113 employees × 4 weeks)

Property	Detail
Time Period	4 weeks of operation
Source Files	week_1.xlsx, week_2.xlsx, week_3.xlsx, week_4.xlsx
Rig Locations	Brass, Ekeremor, Nembe, Southern Ijaw
Total Employees	113 unique employees across all locations
Granularity	One record per employee per week

Field Definitions & Business Meaning

Field	Type	Business Meaning
Employee ID	Integer	Unique identifier for each worker. Enables tracking of individual performance across weeks and ensures records can be matched across files without duplication.
Hours Worked	Numeric	The number of hours the employee worked that week. Used to calculate UPH and to assess the relationship between working hours and productivity.
Units Produced	Integer	The total number of production units the employee contributed to that week. The primary output measure — summed by location to produce total output figures.
Quality Issues	Integer	The number of quality defects or failures recorded for that employee that week. Combined with units produced to calculate UPI — the quality reliability metric.
Production Cost (■)	Numeric	The total cost incurred by that employee's production activity in Naira. Divided by units produced to calculate CPU — the cost efficiency metric.
Training Hours	Integer	The number of hours the employee spent in training that week. Target is at least 1 hour per week. The key independent variable in the training impact analysis.
Rig Location	Categorical	The name of the rig location where the employee is based. The primary grouping variable for all location-level analysis.

6. Data Preparation & Modeling

The four weekly files were structured identically, making combination straightforward — but several preparation decisions were necessary to ensure the model was analytically sound.

Step	Action	Why It Was Necessary
Data Append	All four weekly files were combined into a single unified dataset	Analysis across weeks requires a single data model — separating the files would prevent aggregation across the full four-week period
Week Column	A Week number field (1–4) was added before appending	Enables time-based filtering and trend analysis across weeks on the dashboard
KPI Measures	UPH, CPU, and UPI were calculated as DAX measures rather than computed columns	Measures recalculate dynamically based on any filter applied — location, week, or employee — making the dashboard truly interactive
Target Benchmarks	Fixed target values were hardcoded as reference measures (UPH=11, CPU=750, UPI=250)	Enables variance calculations and conditional formatting to immediately flag underperforming locations
Variance Calculations	Percentage variance from target was calculated for each KPI at each location	Converts absolute numbers into actionable signals — a ■148 cost overrun means less than -19.75% without context
Correlation Measures	Pearson correlation coefficients were calculated between training hours and each KPI, and between hours worked and each KPI	Quantifies the statistical relationship between inputs and outputs — moving management from intuition to evidence
Location Grouping	Data was grouped by Rig Location for all aggregate measures	The primary analytical unit of interest is the location — individual employee data rolls up to location-level performance

7. Management Questions Answered

The business brief raised nine specific questions. Each is answered directly below.

Q1. What is the total number of units produced so far?

285,124	**■220,395,000**	**1,557**	**113**
Total Units Produced	Total Production Cost	Total Quality Issues	Total Employees

Weeks 1–4 Combined	Across All Locations	Across All Locations	4 Rig Locations

The operation has produced 285,124 units in its first four weeks. Brass is the highest-producing location at 101,403 units, reflecting both its larger workforce (39 employees) and its longer average hours. Southern Ijaw is the lowest at 43,990 units — a function of both its smaller team (20 employees) and lower average hours worked.

Q2. What does the cost of production look like?

Location	Total Production Cost	Total Units	CPU Achieved	Target	Status
Brass	■66,515,000	101,403	■717	■750	■ MET
Ekeremor	■55,270,000	54,764	■1,009	■750	■ MISSED by 34.5%
Nembe	■58,560,000	84,967	■689	■750	■ MET
Southern Ijaw	■40,050,000	43,990	■910	■750	■ MISSED by 21.3%
Total / Overall	■220,395,000	285,124	■773	■750	■ MISSED by 3.1%

Only two locations — Brass and Nembe — are producing within the ■750 cost ceiling. Ekeremor's CPU of ■1,009 is the most alarming: at its current production volume, the location is generating a cost overrun of ■259 per unit — equivalent to approximately ■14.2 million in excess spending over four weeks on this location alone.

Q3 & Q4. Employees and average working hours across all locations

Location	Employees	Avg Hours/Week	Total Hours	% Above/Below Avg
Brass	39	63.3 hrs	Highest	+11.6% above network avg
Ekeremor	29	47.6 hrs	Lowest	-16.1% below network avg
Nembe	34	61.5 hrs	Second highest	+8.3% above network avg
Southern Ijaw	20	49.9 hrs	Second lowest	-12.1% below network avg
Network Average	28.25	56.8 hrs	—	Baseline

The two locations with the highest cost overruns — Ekeremor and Southern Ijaw — are also the locations with the lowest average working hours. This is not necessarily a coincidence, and the correlation analysis in Section 8 explores this relationship in more depth.

Q5 & Q6. Average production rate per hour and cost per unit

Location	UPH Achieved	UPH Target	UPH Status	CPU Achieved	CPU Target	CPU Status
Brass	11.20	11	■ +1.8%	■717	■750	■ Under budget
Ekeremor	10.95	11	■ -0.5%	■1,009	■750	■ +34.5% over
Nembe	11.15	11	■ +1.3%	■689	■750	■ Under budget
S. Ijaw	11.03	11	■ +0.3%	■910	■750	■ +21.3% over
Overall	11.02	11	■ +0.2%	■773	■750	■ +3.1% over

Ekeremor is the only location failing on both UPH and CPU simultaneously — producing fewer units per hour while spending more to produce each one. This combination of low productivity and high cost makes it the highest-priority location for operational intervention.

Q7. Locations producing fewer than 250 units before a quality issue

Location	UPI Achieved	UPI Target	Status	Units Before Issue
Brass	246.7	250	■ MISSED — 1.3% below target	~247 units per defect
Ekeremor	122.2	250	■ CRITICAL — 51.1% below target	~122 units per defect
Nembe	209.8	250	■ MISSED — 16.1% below target	~210 units per defect
Southern Ijaw	150.1	250	■ MISSED — 40.0% below target	~150 units per defect

Every single rig location is failing the quality target. This is the most widespread performance failure in the dataset. Ekeremor is in a critical position — producing a quality defect every 122 units, against a target of 250. That means quality incidents are occurring more than twice as frequently as management expects. Even the best-performing location, Brass, is marginally below the 250-unit standard.

Q8. Locations not meeting the 1-hour training target

Location	Avg Training Hrs/Week	Target	Status	Gap
Brass	1.55 hrs	1.0 hr	■ MET — 55% above target	+0.55 hrs above minimum
Ekeremor	0.76 hrs	1.0 hr	■ MISSED — 24% below target	-0.24 hrs below minimum
Nembe	1.18 hrs	1.0 hr	■ MET — 18% above target	+0.18 hrs above minimum
Southern Ijaw	0.69 hrs	1.0 hr	■ MISSED — 31% below target	-0.31 hrs below minimum
Network Average	1.11 hrs	1.0 hr	■ MET — overall	+0.11 hrs above minimum

Ekeremor and Southern Ijaw — the two lowest-performing locations on cost and quality — are the same two locations falling short of the training target. This is a pattern that cannot be dismissed as coincidence, and the next section quantifies exactly how strong that relationship is.

8. The Impact of Training & Working Hours on Performance

Question 9 from the business brief asked whether there is a relationship between training hours and key performance metrics. The correlation analysis below provides a definitive answer — and it is one of the most important findings in this report.

8.1 Training Hours vs Performance Metrics

Metric Pair	Correlation (r)	Strength	Direction	Interpretation
Training Hrs → UPH	0.48	Moderate Positive	↑ More training = higher UPH	Employees with more training tend to produce more units per hour
Training Hrs → CPU	-0.66	Moderate-Strong Negative	↑ More training = lower CPU	Better-trained employees produce units at lower cost — the training is paying back

Metric Pair	Correlation (r)	Strength	Direction	Interpretation
Training Hrs → UPI	0.79	Strong Positive	↑ More training = better quality	Training has the strongest measurable impact on quality — reducing defect frequency significantly

The relationship between training hours and UPI (r = 0.79) is the strongest statistical signal in the entire dataset. At a time when every single rig location is failing the quality target, this finding tells management exactly where to focus: getting Ekeremor and Southern Ijaw — both below the 1-hour training threshold — to meet the training minimum should be the first operational priority.

8.2 Hours Worked vs Performance Metrics

Metric Pair	Correlation (r)	Strength	Direction	Interpretation
Hours Worked → UPH	0.47	Moderate Positive	↑ More hours = higher UPH	Employees who work more hours tend to produce at a higher hourly rate
Hours Worked → CPU	-0.88	Very Strong Negative	↑ More hours = much lower CPU	The strongest relationship in the dataset — working hours drive cost efficiency most powerfully
Hours Worked → UPI	0.80	Strong Positive	↑ More hours = better quality	Employees working more hours maintain higher quality standards throughout their production

The Hours Worked → CPU correlation of -0.88 is the strongest relationship in the entire analysis. It suggests that the high cost-per-unit at Ekeremor and Southern Ijaw is directly connected to the shorter working hours at those locations. Employees who work fewer hours may be producing the same volume, but the fixed overhead costs are being spread over fewer production hours — driving up the unit cost significantly.

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9. Key Insights

Insight 1 — The network is meeting production volume but missing on cost and quality

An 11.02 UPH against a target of 11 looks like a win. But it is a narrow win that masks two more significant failures. The cost of producing those units is 19.75% above target, and the quality standard is being missed at every single location. Management should resist the temptation to declare production a success — the volume of units produced is only one of four performance dimensions, and it is the only one currently green.

Insight 2 — Ekeremor is the network's most critical underperformer

Of the four locations, Ekeremor fails on every measurable dimension: it is the only location missing the UPH target, it has the highest CPU at ■1,009 per unit (34.5% above the ceiling), it has the worst UPI at 122 units per defect, and its staff average only 0.76 training hours per week against a minimum of 1 hour. This is not a location with one weak area — it is a location with a systemic performance problem that requires comprehensive intervention, not incremental adjustment.

Insight 3 — Brass demonstrates that all four targets are achievable simultaneously

Brass is the only location meeting UPH, CPU, and training targets. Its UPI of 246.7 falls just 1.3% short of the quality standard — making it the closest location to full compliance. With 1.55 average training hours per week and 63.3 average working hours, Brass demonstrates that the targets management has set are not unrealistic. They are achievable with the right workforce investment. The question for the other three locations is not whether the targets can be met — it is what changes are needed to meet them.

Insight 4 — Training is the highest-leverage intervention available

With a correlation of 0.79 between training hours and UPI, the data strongly suggests that increasing training at the two non-compliant locations — Ekeremor and Southern Ijaw — would produce measurable quality improvements. This is particularly significant because training is largely a scheduling and commitment decision. It does not require capital investment, procurement, or equipment changes. If management wants the fastest, most cost-effective route to improving quality across the network, enforcing the training minimum at every location is the place to start.

Insight 5 — Southern Ijaw's cost problem may be a scale problem

Southern Ijaw has 20 employees — the smallest workforce of the four locations. Its CPU of ■910 could reflect the fact that fixed costs are being spread over a smaller production base. If this is the case, the intervention is not necessarily about changing how employees work — it may be about right-sizing the workforce or consolidating production activities to achieve economies of scale.

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10. Business Recommendations

Recommendation 1: Enforce the 1-hour training minimum at Ekeremor and Southern Ijaw immediately

Both locations are below the training target. Given the 0.79 correlation between training hours and quality performance, this is the single intervention most likely to produce measurable improvements in UPI across the network — and it requires no capital expenditure, only scheduling discipline.

> *Expected Impact: Quality improvements at Ekeremor and Southern Ijaw within 2–4 weeks. If the training-quality relationship holds, closing the training gap at Ekeremor could meaningfully increase UPI from 122 toward the 250 target.*

Recommendation 2: Commission a cost audit at Ekeremor — the ■1,009 CPU requires immediate investigation

A 34.5% cost overrun at Ekeremor is not a rounding error — it is a structural problem. At the current production rate, Ekeremor is generating an estimated ■14.2 million in excess costs over four weeks. A full operational audit should examine staffing efficiency, material wastage, equipment utilisation, and overhead allocation at this location.

> *Expected Impact: Identification of the primary cost driver at Ekeremor and a path to reducing CPU toward the ■750 ceiling, potentially recovering millions in avoidable spend.*

Recommendation 3: Study and replicate Brass's operational model across all locations

Brass is the only location meeting three of four targets and nearly meeting the fourth. Its combination of higher training hours, longer working hours, and strong cost discipline is producing measurably better results. Operations management should document Brass's practices — scheduling, training protocols, supervision structure — and assess which elements can be transferred to Ekeremor and Southern Ijaw.

> *Expected Impact: A structured framework for performance improvement that is grounded in practices already proven to work within AXZ's own operations.*

Recommendation 4: Investigate whether Southern Ijaw's workforce is appropriately sized

With 20 employees — less than half of Brass's 39 — Southern Ijaw may be suffering from a fixed cost structure that cannot be supported at its current production scale. Management should model whether adding 5–8 employees to Southern Ijaw would reduce its CPU by spreading fixed costs more effectively, or whether consolidating some of its production activities with a nearby location would produce better economics.

> *Expected Impact: A CPU reduction at Southern Ijaw from ■910 toward the ■750 target, and a more defensible long-term cost structure at the smallest rig location.*

Recommendation 5: Set location-specific improvement targets and review performance weekly

A single network-level target creates perverse incentives — a high-performing location like Brass can mask the severity of underperformance at Ekeremor in the aggregate. Management should set formal location-level targets and conduct structured weekly performance reviews for each rig. Locations that miss targets two consecutive weeks should trigger an automatic operational review.

> *Expected Impact: Earlier detection of performance deterioration and faster corrective action — reducing the period during which a location can operate below standard without triggering a formal response.*

11. Expected Outcomes

Recommendation	Expected Outcome
Enforce training minimum at Ekeremor & S.Ijaw	Quality improvements within weeks — UPI moving toward the 250 target at the two worst-performing locations based on the 0.79 training-quality correlation.
Cost audit at Ekeremor	Root cause of ■1,009 CPU identified and a cost reduction roadmap established — potential recovery of ■14M+ in avoidable spend.
Replicate Brass operational model	A transferable performance framework grounded in proven in-house practices, reducing reliance on external benchmarks.
Southern Ijaw workforce review	CPU reduction toward ■750 target through either workforce expansion or activity consolidation.
Weekly location-level performance reviews	Earlier identification of emerging underperformance — reducing the time a location can operate below standard from weeks to days.

12. Dashboard Design

A four-page Power BI dashboard was built to answer all nine management questions in a single, interactive interface. Each page addresses a specific analytical dimension — from operational overview to correlation analysis to individual rig-level detail.

Page	Title	Business Purpose	Key Visuals
Page 0	Operational Overview	Answers questions 1–8 — provides a complete summary of all four KPIs by location with employee and training context	KPI cards, location bar charts (UPH, CPU, UPI, Units, Cost, Issues), training and hours charts, employee counts. Hour-range filter (20–40, 41–60, 61–80, 81–100 hrs) for segmented analysis.
Page 1	Training Hours Impact	Answers question 9 — shows correlation between training hours and UPH, CPU, and UPI	Three gauge charts showing correlation coefficients (-1 to +1) and three scatter plots visualising the training-performance relationship for each KPI
Page 2	Hours Worked Impact	Extends question 9 — shows correlation between hours worked and the same three KPIs	Three gauge charts and scatter plots for Hours Worked vs UPH, CPU, and UPI — revealing the -0.88 cost correlation
Page 3	Rig-Level Detail Table	Provides drill-down to individual rig numbers within each location for granular investigation	Full data table with UPH, CPU, UPI, Total Units, Total Cost, and Quality Issues by rig and location with variance percentages

13. Limitations

- Four weeks of data is a short observation period. Performance patterns may shift significantly as operations mature — particularly as training hours accumulate and employees gain experience.
- The analysis is descriptive, not causal. A strong correlation between training hours and UPI (r = 0.79) does not prove that training causes quality improvements — it establishes a pattern that warrants further investigation and investment.
- Individual employee records are aggregated to location level for all KPI analysis. High performers within a poorly-performing location — or vice versa — are not visible in the summary metrics.
- The dataset does not include external factors such as equipment age, maintenance schedules, material quality differences, or environmental conditions that may contribute to performance variation across locations.
- No financial model is available to translate the ■14M estimated cost overrun at Ekeremor into margin impact or profitability implications.

14. Future Improvements

Extend the dataset to 8–12 weeks

A longer observation period would validate whether the training-quality correlation strengthens or weakens over time, and allow trend analysis to distinguish between improving, stable, and deteriorating locations.

Build a predictive performance model

Using the correlation relationships identified in this analysis, a regression model could be built to predict expected KPI outcomes based on training hours and working hours — enabling management to forecast performance rather than simply measure it.

Individual employee performance tracking

Adding an employee-level dashboard page would surface high performers who could serve as peer coaches within underperforming locations, and identify employees who may need targeted support.

Integrate financial P&L; data

Connecting the production cost data to revenue and margin information would allow the dashboard to show not just cost per unit, but profit per unit and location-level profitability — giving management a complete financial picture.

Automated weekly performance alerts

Building Power BI data alerts to notify managers when any location falls below target on any KPI would compress the detection-to-response cycle from weekly reviews to real-time notifications.

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15. Technical Skills Demonstrated

Skill	Application in This Project
Power Query (M Language)	Combined four weekly Excel files into a single unified data model, added a Week dimension field, and validated data integrity across all 452 records before loading.
DAX Measures	Created dynamic KPI measures (UPH, CPU, UPI) that recalculate on any filter, target reference measures (UPH Target, CPU Target, UPI Target), variance calculations, and Pearson correlation coefficient measures for the training and hours worked analysis.
Power BI Dashboard Design	Designed a four-page interactive dashboard with hour-range filters, KPI cards with conditional formatting, location-level bar charts, scatter plots for correlation visualisation, and gauge charts for correlation coefficient display.

Skill	Application in This Project
Correlation Analysis	Quantified the statistical relationship between training hours and all three KPIs, and between hours worked and all three KPIs — translating a management intuition into a statistically supported finding.
Business Translation	Converted raw production numbers into variance percentages, location rankings, cost overrun estimates, and actionable recommendations — moving from data description to business intelligence.

16. Conclusion

Four weeks into operations, AXZ Oil and Gas is producing units at the right rate but at the wrong cost and the wrong quality level. The headline UPH metric creates a misleading impression of overall health — one that dissolves the moment the cost and quality numbers are examined.

The data points clearly to where the problems are concentrated. Ekeremor accounts for the worst performance on every metric. Southern Ijaw has the second-most severe cost problem. Both locations are also the ones failing the training target. That alignment is not a coincidence — the correlation analysis confirms that training hours and working hours are both statistically linked to quality and cost outcomes across the network.

The good news is that Brass — meeting three of four targets in its first four weeks — demonstrates that the performance standards management has set are attainable. The path forward is not to lower the targets. It is to understand what Brass is doing and systematically bring the other three locations up to the same standard. Starting with training compliance at Ekeremor and Southern Ijaw is the fastest and most cost-effective first step available.

Sign-Off

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Client: AXZ Oil and Gas Production Company

Commissioned by: SkillAhead Analytics

Tool: Power BI | Dataset: Weeks 1–4 (452 records)

Production is a process. Performance is a choice.